Chapter 14: ADM Techniques

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 14 (pp. 589-668)

Overview

Chapter 14 (pp. 589-668) covers the eight "ADM Techniques" that the Standard has split into their own document because they are used at many points in the ADM (p. 589). Section 14.1 asks which of them are really techniques (pp. 589-592). The sections then cover Principles, Enterprise vs. Architecture (14.2, pp. 592-601); Stakeholder Management, with analysis, mapping, Communications Plan, and engagement (14.3, pp. 601-615); Architecture Patterns (14.4, pp. 616-617); Gap Analysis (14.5, pp. 617-624); Interoperability Requirements (14.6, pp. 624-631); the Business Transformation Readiness Assessment (14.7, pp. 631-640); Risk Management, with classification, identification, assessment, mitigation, monitoring, and visualization (14.8, pp. 640-659); and Architecture Alternatives and Trade-Offs (14.9, pp. 659-666). Section 14.10 summarizes (pp. 666-668). Leave with this idea: these are the reusable tools that the ADM phases keep calling on. In Chapter 12, Phase A uses stakeholder analysis and readiness, B-D use gap analysis and trade-offs, and E-H use risk management (pp. 590-591). Chapter 14 is part of the GREEN and BLUE reading tours (p. 589).

> **Reading note:** The eight techniques, the principle structure and five quality criteria, the Power Grid, the gap matrix, the readiness factors, and the risk approach come from the Standard. Jager's views: several "techniques" are not really techniques (pp. 590-591); the "Enterprise Architecture Principles" term in a Series Guide is an inconsistency (p. 597); a Communications Plan should reach the whole organization (pp. 611-612). Keep apart **Enterprise Principles** (organization-wide, board-approved) and **Architecture Principles** (architecture work, Architecture Board-approved).

Detailed Section Breakdown

Chapter Opener (p. 589)

- **Unbundled topics.** Principles, Stakeholder Management, and Risk Management, plus patterns, gap analysis, interoperability, readiness, and trade-offs, were split out because they are invoked at several points. This adds to the Standard's modularity (p. 589).

14.1 Technique or No Technique (pp. 589-592)

- **What TOGAF means by "technique".** Not always a specific method, but an elaboration of a frequently used topic that needs explanation to be applied properly (p. 590).

The Standard's eight techniques	Short description (p. 591)
Architecture Principles	Rules and guidelines for consistent, coherent architecture and governance
Stakeholder Management	Identify, analyze, and manage stakeholders
Architecture Patterns	Standardized solutions to common problems
Gap Analysis	Differences between Baseline and Target
Interoperability Requirements	Determining interoperability requirements
Business Transformation Readiness Assessment	Identifying business transformation issues

The Standard's eight techniques	Short description (p. 591)
Risk Management	Identify and mitigate issues before they hit the architecture
Architecture Alternatives and Trade-Offs	Identify alternative targets and trade off between them

Jager's commentary (author's view): Stakeholder Management and Risk Management are essential but "not techniques", otherwise roadmapping would be one too. A readiness assessment and a gap analysis *are* techniques, because they complement an existing activity (Phases A and E, and Phases B-D respectively) (pp. 590-591).

14.2 Principles (pp. 592-601)

Aspect	Enterprise Principles	Architecture Principles
Scope	Organization-wide, business and IT	Focused on Enterprise Architecture
Purpose	Guide overall decisions and strategy	Framework for architecture design and execution
Application	Broad, influences policy and culture	More technical, used in architecture development
Examples	Agility, customer focus, security	Reusability, scalability, standardization

(Table 14-1, p. 593)

14.2.1 The Added Value of Principles (pp. 593-594)

- Clarity, consistency, and direction. Strategic alignment of business and IT. Standardized approaches. A structured way to evaluate options and trade-offs. Accountability. Shared understanding. Regulatory compliance. Less complexity and duplication (pp. 593-594).

14.2.2 The Structure of a Principle (pp. 594-597)

Element	What it contains (Table 14-2)	Tips
Name	A short, clear title	Capture the essence. No product names, no ambiguous words
Statement	A concise declaration of intent	Convey the rule without ambiguity
Rationale	Why it matters (business value)	Mention consistency with or precedence over other principles
Implications	Practical impact, trade-offs, required actions	Requirements in resources, time, money, activities: "How does this affect the business?"

(pp. 594-595)

Five quality criteria for a set of principles	Meaning
Understandable	Quickly grasped. Intent clear, minimizing violations
Robust	Enables high-quality decisions and enforceable policies, even in controversial cases
Complete	Covers every potentially important situation
Consistent	Balanced interpretation possible. No principle violates the spirit of another
Stable	Enduring, but with an amendment process

(pp. 595-596)

- **Storage.** Enterprise and Architecture Principles live in the Governance Repository of the Architecture Repository (Figure 14-1, p. 596).

Jager's commentary (author's view): The Series Guide "Using the TOGAF Standard in the Digital Enterprise" uses "Enterprise Architecture Principles", which blurs two distinct types. In practice, most organizations' "principles" come from IT and often look like requirements (p. 597).

14.2.3 Enterprise Principles (pp. 597-599)

- "Provide a basis for decision-making throughout an enterprise and inform how the organization sets about fulfilling its mission". Ideally drafted by the board. In reality drafted by the Enterprise Architect and approved by the board, with stakeholder buy-in. Departments may have their own sets, but these must be aligned (pp. 597-598).
- Example: *Comply with laws and regulations* (Table 14-3). More examples are in the Principles Catalog, Table 12-5 (pp. 598-599).

14.2.4 Architecture Principles (pp. 599-601)

- Guide the architecture process and the development, maintenance, and use of the EA. Drafted by the Enterprise Architect with key stakeholders and approved by the **Architecture Board** (p. 599).
- **Hierarchy (Figure 14-2).** Segment principles (business, data, application, technology) are informed by and bounded by Enterprise Principles (pp. 599-600).
- Example: *Common vocabulary and definitions* (Table 14-4). Each Architecture Principle should link explicitly to business goals and architectural drivers (pp. 600-601).

14.3 Stakeholder Management (pp. 601-615)

- **Definition.** "The process of identifying and understanding the needs, interests, and expectations of an organization's various stakeholders and planning and implementing actions to address those interests" (p. 601).
- **Stakeholder.** "Any individual, group, or organization that can affect, be affected by, or perceive themselves to be affected by" an EA initiative (pp. 602-603).
- **Why it matters.** Stakeholders give input and requirements, raise concerns, influence decisions, assess impacts (their views, Ch. 7.6), and underpin governance. Early identification of powerful stakeholders secures resources and support (pp. 601-603).
- **When.** Start in Phase A and update in every phase, including E, F, and H (p. 603).

Step (Figure 14-3)	Part of
1. Identify stakeholders	Stakeholder analysis
2. Analyze and classify stakeholder positions	Stakeholder analysis
3. Plan communications	Communications Plan
4. Engage stakeholders	Stakeholder engagement
5. Monitor and review	Stakeholder engagement

(pp. 604-605)

14.3.1 Stakeholder Analysis (pp. 605-609)

- **Stakeholder Analysis Model (Table 14-5).** Per stakeholder: ability to disrupt change (power), current and required understanding, current and required commitment, required support (pp. 606-607).
- **Stakeholder Power Grid (Figure 14-4):**

Quadrant	Power / interest	Communication	Example
Key Players	High / high	Most frequent	Data and process owners
Keep Satisfied	High / low	Slightly less often	Most C-level roles
Keep Informed	Low / high	Regular email or meetings	Program manager
Minimal Effort	Low / low	Least, but may still matter	Employees

(pp. 607-608)

- In the example (Table 14-6): CEO is Keep Satisfied, CIO is Key Player, CFO is Keep Informed, HR Manager is Minimal Effort. Each classification sets the communication frequency and method used in the Communications Plan (p. 609).

14.3.2 Stakeholder Mapping (pp. 609-611)

- The **Stakeholder Catalog** (Table 14-7) adds key concerns and the artifacts of interest to each stakeholder (e.g., the CFO is interested in the Application and Technology Portfolio Catalogs) (pp. 609-611).

14.3.3 Communications Plan (pp. 611-614)

- A critical success factor. Per stakeholder group: purpose, information needed, frequency, method (Table 14-8: CxO fortnightly meetings; data and process owners monthly; employees email plus meetings) (pp. 611-614).

Jager's commentary (author's view): The Standard limits the plan to key stakeholders, but it should reach the whole organization. It should explain everyone's role in implementing the EA and schedule organization-wide progress updates, because transparency builds acceptance (pp. 611-612).

14.3.4 Stakeholder Engagement (p. 615)

- Go beyond the static plan with appealing visuals, infographics, posters, and themed rooms. Invite input. Regularly evaluate effectiveness and update the analysis when new stakeholders appear (p. 615).

14.4 Architecture Patterns (pp. 616-617)

- **Patterns put building blocks in context.** "Building blocks are what you use; patterns can tell you how you use them, when, why, and what trade-offs you have to make". A pattern combines proven ABBs and SBBs (Figure 14-5) (pp. 616-617).
- **Not design patterns.** They are often confused with software design patterns (p. 617).
- **Not yet integrated** into the ADM by the Standard, but they can be defined in B-D when reusable blocks are identified (p. 617).

14.5 Gap Analysis (pp. 617-622)

- **Definition.** Identifies the difference between Baseline and Target (capabilities, processes, applications, technology). "Highlight a shortfall... items that have been deliberately omitted, accidentally left out, or not yet defined" (p. 617).
- **Basic steps.** Identify baseline and target; compare; categorize gaps (missing functionality, technology gaps, process inefficiencies); group activities into work packages; prioritize by impact and feasibility (p. 618).
- **Hospital example (Figures 14-6, 14-7).** Application A (baseline) is replaced by Application B (target), which automates the missing Inflow Support capability and also Personnel Information Management. Gap A (baseline to transition) and Gap B (transition to target) each get a work package. Gaps are found mainly in B-D, and the real analysis and work packaging happen in E (pp. 618-620).

Gap Analysis Matrix steps	Instruction
Set up the matrix	Rows = baseline ABBs, columns = target ABBs
Add New and Eliminated	An extra bottom row "New" and an extra right column "Eliminated"

Gap Analysis Matrix steps	Instruction
Identify common blocks	Mark "Included" where an ABB is in both
Review missing blocks	Baseline-only ABBs go in Eliminated. Check whether this is correct or an accidental omission to reinstate
Document new blocks	Target-only ABBs go in the New row, with implementation steps

(pp. 621-622, Table 14-9)

- Everything under New or Eliminated is a gap: either explained as correctly eliminated, or to be addressed by work packages that feed the Architecture Roadmap (Ch. 12.8.3) (p. 622).

14.6 Interoperability Requirements (pp. 623-627)

- **Idea.** Compare entities and decide the form and degree of interoperability needed. Interoperability is "the ability to share information and services" (p. 623).
- **Cross-mapping.** E.g., stakeholder-to-stakeholder (built in Phase B, Table 14-10) or stakeholder-to-application (Table 14-11). The phase in which a mapping is created depends on its entities (pp. 623-625).

Degree	Name	Meaning
1	Unstructured Data Exchange	Human-interpretable free text (estimates, papers)
2	Structured Data Exchange	Human-interpretable structured data, with manual compilation or dispatch
3	Seamless Sharing of Data	Automated exchange between systems based on a common exchange model
4	Seamless Sharing of Information	Universal interpretation of information through cooperating applications

(p. 624)

- **Sub-degrees.** A letter refines a degree (e.g., 2A, 3C, 4D) to express standardization or compatibility (Figure 14-8, p. 625).
- **Appears across the ADM** (Figure 14-9). Benefit: better integration and fewer data silos (pp. 626-627).

> **Text note:** p. 626 describes the 2A cell as containing "the letter B" (Common Data Exchange). The table shows 2A, so treat the letter in the text as a slip.

14.7 Business Transformation Readiness Assessment (pp. 627-632)

- **Why.** Architecture change affects workforce, culture, and skills. A change-averse culture can sink a good design. The assessment is done collaboratively by HR, lines of business, and (IT) planners, for successful implementation in E and F (pp. 627-628).

Step	What to do	Page
Identify key readiness factors	e.g., cultural readiness, workforce skills, leadership alignment, change capability	p. 628
Use maturity models	Assess baseline and target per factor (Ch. 15 format)	p. 628
Determine readiness factor ratings	Current maturity and gaps	p. 628
Identify and mitigate risks	Using Risk Management (14.8)	p. 628
Integrate into the Implementation and Migration Plan	Created in E, finalized in F	p. 628

- **Example (Tables 14-12, 14-13).** Factors are scored 1-5 (poor, fair, acceptable, good, excellent): leadership commitment, employee engagement, organizational culture, skills and competencies, Risk Management, stakeholder alignment. Each is then rated by *urgency*, *readiness status*, and *remediation difficulty* to give management a quick overview (pp. 629-630).
- **Monitor in Phase G** and take corrective action through the Governance Framework (p. 630).

14.8 Risk Management (pp. 632-656)

- **Phases where it is addressed (Figure 14-10).** A (initial high-level risks, recorded in the Statement of Architecture Work); E (implementation risks from gaps, influencing approach choice); F (detailed migration risks from the Consolidated Matrix, plus fallback plans); G (risks during implementation, governance oversight); H (continuous monitoring of changes) (pp. 632-634).
- **Not formally addressed** in Preliminary or B-D, although B-D risks surface via gaps and are recorded in Phase E (p. 634).
- **TOGAF vs. Enterprise Risk Management.** TOGAF's technique is a simplified version focused on *architecture project* risk. ERM (Series Guide G152) also covers business, system, information, privacy, compliance, and change risk (p. 635).
- **Initial vs. residual risk (Figure 14-11).** Initial = new, before mitigation, so apply the technique. Residual = what remains after mitigation, usually already accepted (pp. 635-636).
- **Five steps.** Classify, identify, assess (impact x frequency), mitigate and assess residual, monitor (in G). Mitigation must truly reduce impact, not just shift the risk to a similarly high quadrant (pp. 636-637).

14.8.1 Risk Classification (pp. 638-642)

- **Primary categories.** Time (schedule), cost (budget), scope (p. 638).
- **Additional categories.** Contractual, technology, scope and complexity, environmental (corporate), people, client acceptance (p. 639).
- **By domain.** Business, information, application, technology risks (p. 640).
- **Implementation Factor Catalog + risk classification (Table 14-14).** E.g., replacing Application A with B is an environmental (corporate) risk and a people risk (training needed, headcount reduction) (pp. 641-642).

14.8.2 Risk Identification (pp. 642-643)

- Risks surface from the B-D gaps and from readiness assessments. They are detailed in the Implementation Factor Catalog (E) and addressed in F. Capability maturity models also reveal risks (p. 642).
- **Risk Management Plan** (templates from PMBOK, PRINCE2): document information, approvals, approach (tools, recorded data, classification metrics, risk tolerance) (pp. 642-643).

14.8.3 Initial Risk Assessment (pp. 644-647)

- Takes place in E and F, scoring impact and frequency on 1-5 scales (p. 644).

Impact (Table 14-15)	Frequency (Table 14-16)
5 Catastrophic: could lead to bankruptcy	5 Frequent: very often or continuously
4 Significant: severe loss across more than one line of business	4 Probable: several times per transformation cycle
3 Moderate: small loss to a line of business, reduced ROI	3 Occasional: sporadically
2 Low: very minor impact, no financial loss	2 Seldom: possible, unlikely more than once
1 Negligible: minimal impact	1 Unlikely: not likely in a cycle

(p. 645)

- **Classifications (Table 14-17).** Critical (C): likely failure with serious consequences. High (H): significant failure of parts. Moderate (M): noticeable failure jeopardizing some goals. Low (L): some goals not fully achieved (p. 646).
- **Classification scheme (Table 14-18).** Impact against frequency gives C/H/M/L (e.g., Catastrophic + Frequent = C; Moderate + Probable = M). This is the basis of the Risk Heat Map (p. 647).
- **The risk is to the organization**, not to the architecture work (p. 646).

14.8.4 Risk Mitigation and Residual Risk Assessment (pp. 647-650)

- Strategies range from monitoring or accepting a risk to full redundancy. Prioritize frequent, high-impact risks (p. 647).
- **Risk Identification and Mitigation Assessment Matrix (Table 14-19).** Columns: risk, preliminary impact, frequency, and level (impact x frequency), mitigation, then residual impact, frequency, and level. E.g., "Reduction in headcount" 3x4=12, after automating capabilities 2x2=4 (pp. 648-649).
- A drop from catastrophic to critical is still extreme, so reconsider such a mitigation (p. 649).

14.8.5 Risk Monitoring (pp. 650-651)

- Residual risks need approval through the Governance Framework (possibly corporate governance). The matrix is a governance artifact maintained in Phase G. Unmitigated critical risks may trigger a new ADM cycle. Phase H monitors risks of changes (pp. 650-651).

14.8.6 Visualizing Risks (pp. 651-656)

- **Risk Heat Map (Figure 14-12).** Frequency on x, impact on y. Green is low, yellow medium, orange/red high or critical. A *risk tolerance* line marks the highest acceptable level. Eyes go to the upper right (pp. 651-653).
- **Calculating risk.** "Risk = Impact (effect) x Frequency (likelihood)". There is no scientific scoring, so use common sense and agreed definitions. Risk #1 (3x4=12) is medium. Risk #2 (2x3=6) is low (pp. 653-654).
- **Adding risks to diagrams.** ArchiMate (not TOGAF) supports risk modelling, e.g., risks added to a Plateau Planning Diagram (Figure 14-14) (p. 655).

14.9 Architecture Alternatives and Trade-Offs (pp. 656-662)

- **Why.** A single solution rarely satisfies all stakeholders. Presenting at least two alternatives reveals hidden agendas and principles (p. 656).
- **Method (Figure 14-15).** Use the vision, principles, requirements, and other information to set criteria. Formulate alternatives. Select one or combine elements. It applies at any stage or level (p. 657).
- **No fixed criteria.** Consider principles, requirements, vision, and stakeholder concerns, discuss pros and cons, and add viewpoints if needed to expose dependencies, risks, and uncertainties (p. 658).

Criteria for alternatives	Question
Flexibility	How adaptable to changing business needs?
Time and cost	Timeframes and costs, including transition and stabilization
Benefit realization	Over what period do the benefits come?
Architectural alignment	Does it follow established styles and guidelines?
Delivery method	Reuse, build, or buy?
Business impact	How little disruption to existing capabilities?
Risk profile	What risks and mitigations?

(p. 659)

Jager's commentary (author's view): In his Lemon-A-de example (from *Getting Started with Enterprise Architecture*), scenario 1 (hire staff and run events) raises brand awareness but risks unmet demand. Scenario 2 (training) raises production but risks unsold inventory. Trade-off analysis often leads to a phased roadmap that eventually delivers both (pp. 659-662).

14.10 Summary (pp. 663-668)

- Techniques are recurring, elaborated topics. Some are not true techniques. There are eight (pp. 663-664).
- Principles, Stakeholder Management (Phase A onward), gap analysis, the five E/F matrices and tables, interoperability, readiness, risk (initial vs. residual, heat maps), and alternatives (pp. 664-668).

Jager's commentary (author's view): Why architecture patterns are in the Standard at all "remains a mystery", because the Standard itself says they are not yet integrated into the ADM (p. 665).

Key Terms & Definitions

Term	Definition	Page
ADM technique	An elaboration of a recurring topic used across the ADM	p. 590
Enterprise Principle	High-level guideline shaping the organization's strategy, culture, and decisions. Board-approved	pp. 592, 597
Architecture Principle	Guideline for architecture work. Architecture Board-approved	pp. 592, 599
Principle structure	Name, Statement, Rationale, Implications	p. 594
Stakeholder	Anyone who can affect, be affected by, or perceive themselves affected by the initiative	pp. 602-603
Stakeholder Power Grid	Power vs. interest: Key Players, Keep Satisfied, Keep Informed, Minimal Effort	pp. 607-608
Communications Plan	Per-group communication goals, methods, frequencies	pp. 611-614
Architecture pattern	A proven combination of ABBs and SBBs that solves a recurring problem	pp. 616-617
Gap Analysis Matrix	Baseline ABBs vs. target ABBs with Included, New, and Eliminated	pp. 621-622
Degrees of interoperability	1 Unstructured, 2 Structured data exchange, 3 Seamless data, 4 Seamless information	p. 624
Readiness factor rating	Urgency, readiness status, remediation difficulty	pp. 629-630
Initial / residual risk	Risk before mitigation / risk remaining after mitigation	pp. 635-636
Risk score	Impact x Frequency	p. 653
Risk tolerance	The highest level of risk still considered acceptable	p. 653

Figures & Tables Referenced

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Tables 14-3, 14-4, Figure 14-2	Example principles. Segment principle sets	pp. 598-600
Figure 14-3	Steps of stakeholder analysis and engagement	p. 604
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Figure / Table	What it shows	Page
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Figures 14-12 to 14-14	Risk Heat Maps. Plateau diagram with risks	pp. 652-655
Figures 14-15 to 14-17	Alternatives and Trade-Offs method and Lemon-A-de example	pp. 657-661

Self-Check Questions

Q: List the eight ADM techniques.

A: Architecture Principles, Stakeholder Management, Architecture Patterns, Gap Analysis, Interoperability Requirements, Business Transformation Readiness Assessment, Risk Management, Architecture Alternatives and Trade-Offs (p. 591).

Q: Who approves Enterprise Principles and who approves Architecture Principles?

A: Enterprise Principles are approved by the board of directors or senior management. Architecture Principles are approved by the Architecture Board (pp. 597-599).

Q: Name the four parts of a principle and the five quality criteria for a set of principles.

A: Parts: Name, Statement, Rationale, Implications. Criteria: Understandable, Robust, Complete, Consistent, Stable (pp. 594-596).

Q: Where in the Architecture Repository are principles stored?

A: In the Governance Repository (p. 596).

Q: Classify these on the Power Grid: high power and low interest; low power and high interest.

A: Keep Satisfied; Keep Informed (pp. 607-608).

Q: What extra element does Jager want in a Communications Plan?

A: It should reach the whole organization: explain everyone's role in implementing the EA and schedule organization-wide progress updates (pp. 611-612).

Q: In a Gap Analysis Matrix, what does an ABB in the Eliminated column mean?

A: It exists in the baseline but not the target. Either it was correctly eliminated, or it is an accidental omission to reinstate in the next iteration (pp. 621-622).

Q: What are the four degrees of interoperability?

A: 1 Unstructured Data Exchange, 2 Structured Data Exchange, 3 Seamless Sharing of Data, 4 Seamless Sharing of Information (p. 624).

Q: In which ADM phases is Risk Management explicitly addressed, and where is it not?

A: Addressed in A, E, F, G, and H. Not formally in the Preliminary Phase or in B-D (B-D risks surface via gaps and are recorded in E) (pp. 632-634).

Q: Calculate the risk score and class for impact 3 (moderate) and frequency 4 (probable).

A: 3 x 4 = 12, a medium (moderate) risk (pp. 647, 654).

Q: Why can a mitigation that reduces a risk from catastrophic to critical still be inadequate?

A: The risk remains extremely high. Mitigation must noticeably reduce the impact, not just move the risk to a similarly high quadrant (pp. 637, 649).

Q: Name four criteria for comparing architecture alternatives.

A: Any four of: flexibility, time and cost, benefit realization, architectural alignment, delivery method, business impact, risk profile (p. 659).

Q: What three ratings are used in the readiness factor rating?

A: Urgency, readiness status (from the 1-5 score), and remediation difficulty (pp. 629-630).